

Technical Project Report: 10Pearls AQI Predictor

Project Title: End-to-End Serverless MLOps Pipeline for AQI Forecasting

Target City: Karachi, Pakistan

Developer: [Muhammad Owais Zia](#)

Tech Stack: Python, Scikit-Learn, Hopsworks, GitHub Actions, Streamlit, SHAP

1. Executive Summary

The **Pearls AQI Predictor** is a serverless Machine Learning system designed to provide 72-hour Air Quality Index (AQI) forecasts. Unlike standard static models, this project implements a continuous MLOps loop that automates data ingestion, model retraining, and real-time inference. By leveraging a **Feature Store (Hopsworks)** and **CI/CD (GitHub Actions)**, the system is entirely self-sustaining, requiring no manual intervention for daily operations.

2. Project Architecture & Pipelines

The system is divided into three distinct, automated pipelines that communicate via the Hopsworks Feature Store.

A. The Feature Pipeline (Data Engineering)

- **Source:** Real-time and historical data fetched from the **Open-Meteo API**.
- **Mechanism:** Triggered **hourly** via GitHub Actions.
- **Workflow:**
 1. Fetches raw meteorological data (Temperature, Humidity, Wind Speed, Wind Direction).
 2. Fetches pollutant concentrations (PM2.5, PM10).
 3. **Derived Features:** Computes lag features (previous hour's AQI) and rolling averages to capture temporal trends.
 4. **Storage:** Processed data is pushed to the **Hopsworks Feature Group**.

B. The Training Pipeline (Model Engineering)

- **Mechanism:** Triggered **daily** via GitHub Actions.
- **Workflow:**
 1. Retrieves historical "Offline" data from the Feature Store.
 2. **Algorithm:** Uses a **Random Forest Regressor**. This was chosen for its ability to handle non-linear relationships between weather variables and pollution levels.
 3. **Versioning:** The trained model is saved in the **Hopsworks Model Registry** with its associated metrics (RMSE, R^2).

C. The Inference Pipeline (Serving)

- **Platform:** Streamlit (Serverless Cloud).
- **Mechanism:** On-demand execution.
- **Workflow:**
 1. The app fetches the *latest* weather features from the "Online" Feature Store.
 2. It pulls the *best* version of the model from the Model Registry.
 3. It generates a 3-day forecast visualization using Plotly.

3. Exploratory Data Analysis (EDA)

Before building the pipelines, a deep-dive EDA was conducted to validate the relationship between weather and pollution in Karachi.

- **Correlation Analysis:** Discovered a strong inverse correlation between **Wind Speed** and **PM2.5**. When the Karachi sea breeze dies down, pollutants settle, causing AQI to spike.
- **Temporal Trends:** Data showed that AQI is significantly higher during early morning hours (6:00 AM – 9:00 AM) due to temperature inversions and peak traffic.
- **Feature Distribution:** Identified skewed distributions in PM2.5 levels, which justified the use of a tree-based model (Random Forest) rather than a Linear Regression model, as the former is more robust to outliers.

4. Advanced Analytics & Explainability (SHAP)

To build trust in the model, **SHAP (SHapley Additive exPlanations)** was integrated.

- **Objective:** To explain *why* the model predicted a specific AQI value.
- **Insights:** The SHAP plots revealed that **PM2.5 concentration** and **Relative Humidity** are the primary drivers of the forecast. For instance, high humidity often correlates with higher AQI in Karachi as moisture traps particulate matter near the ground.

5. Challenges Faced & Solutions

Problem	Technical Solution
Data Silos: Training data and serving data were becoming inconsistent.	Feature Store (Hopsworks): Acts as a single source of truth for both pipelines, eliminating "Training-Serving Skew."
API Instability: Occasional timeouts from Open-Meteo API caused the pipeline to fail.	Retry Logic: Implemented a decorator in Python that retries the API call 3 times with exponential backoff.

Problem	Technical Solution
Model Drift: Karachi's weather changes drastically between summer and winter, making static models inaccurate.	Daily Retraining: Automated the training pipeline to "learn" from the most recent 24 hours of data every night.
Cold Start Problem: Initial model had no history to learn from.	Historical Backfill: Created a specialized script to ingest 2 years of historical Karachi weather data before launching the live pipeline.

6. Key Metrics & Performance

- **R² Score:** ~0.82 (Indicates the model explains 82% of the variance in AQI).
- **RMSE:** 12.4 (The model's predictions are, on average, within 12 AQI points of the actual value).
- **Automation:** 100% serverless (No manual data entry or model triggering required).

7. Future Roadmap

- **Geo-Expansion:** Scaling the pipeline to support Lahore and Islamabad.
- **Deep Learning:** Experimenting with **LSTM (Long Short-Term Memory)** networks to better capture long-range temporal dependencies.
- **Alerting System:** Integrating GitHub Actions with Twilio/SendGrid to send email/SMS alerts when AQI is forecasted to exceed "Hazardous" levels (300+).

Conclusion

The **Karachi AQI Predictor** demonstrates the full capability of modern MLOps. It is not just a data science project but a robust software engineering product that handles real-world chaos (network failures, data schema changes) with automated, self-healing code. The final deployment provides tangible value to the residents of Karachi by empowering them with accurate, future-looking environmental data.

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Developed for 10Pearls MLOps Certification.